質点系 … たがいに力を及ぼしあう複数の質点から構成されるもの.

内力と外力の違い … 質点系を構成する質点どうしが及ぼしあう力が「内力」
質点系以外から加わる力が「外力」.

剛体 … どんなに力を加えても形が変わらない物体.

練習問題1

16. (1) $V = \dfrac{kx^2 + ky^2}{2}$

$$F_x = -\frac{\partial V}{\partial x} = -\frac{2kx}{2} = -kx$$

$$F_y = -\frac{\partial V}{\partial y} = -\frac{2ky}{2} = -ky$$

(2) $V = \dfrac{q_1 q_2}{4\pi\varepsilon_0 r}$ $\quad r = \sqrt{x^2+y^2+z^2}$

$$F_x = -\frac{\partial V}{\partial x} = -\frac{dV}{dr}\frac{\partial r}{\partial x} = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}\cdot\frac{x}{\sqrt{x^2+y^2+z^2}} = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}\cdot\frac{x}{r}$$

$$F_y = -\frac{\partial V}{\partial y} = -\frac{dV}{dr}\frac{\partial r}{\partial x} = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}\cdot\frac{y}{r}$$

$$F_z = -\frac{\partial V}{\partial z} = -\frac{dV}{dr}\frac{\partial r}{\partial z} = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}\cdot\frac{z}{r}$$

(3) $V = C\dfrac{e^{-kr}}{r}$ $\quad r = \sqrt{x^2+y^2+z^2}$

$$F_x = -\frac{\partial V}{\partial x} = -\frac{dV}{dr}\cdot\frac{\partial r}{\partial x} = -\left(-\frac{Ce^{-kr}}{r^2} - \frac{Cke^{-kr}}{r}\right)\cdot\frac{x}{r} = C\left(\frac{1}{r^2}+\frac{k}{r}\right)e^{-kr}\cdot\frac{x}{r}$$

$$F_y = -\frac{\partial V}{\partial y} = -\frac{dV}{dr}\cdot\frac{\partial r}{\partial y} = C\left(\frac{1}{r^2}+\frac{k}{r}\right)e^{-kr}\cdot\frac{y}{r}$$

$$F_z = -\frac{\partial V}{\partial z} = -\frac{dV}{dr}\cdot\frac{\partial r}{\partial z} = C\left(\frac{1}{r^2}+\frac{k}{r}\right)e^{-kr}\cdot\frac{z}{r}$$

宿題2

$$\begin{cases} m \cdot v_0 = MV + mv & \cdots ① \\ \dfrac{m \cdot v_0^2}{2} = \dfrac{MV^2}{2} + \dfrac{mv^2}{2} & \cdots ② \end{cases}$$

①より

$$v = \frac{mv_0 - MV}{m} \quad \cdots ③$$

これを②に代入して、

$$mv_0^2 = MV^2 + m\left(\frac{mv_0 - MV}{m}\right)^2$$

$$= MV^2 + m\left(v_0^2 - \frac{2MVv_0}{m} + \frac{M^2V^2}{m^2}\right)$$

$$= MV^2 + mv_0^2 - 2MVv_0 + \frac{M^2V^2}{m}$$

$$0 = MV^2 - 2MVv_0 + \frac{M^2V^2}{m}$$

$$0 = V^2\left(M + \frac{M^2}{m}\right) - 2MVv_0$$

$$0 = V\left(V\left(M + \frac{M^2}{m}\right) - 2Mv_0\right)$$

$$\therefore V = 0 \text{ または } V\left(\frac{mM + M^2}{m}\right) = 2Mv_0$$

$$V = \frac{2Mv_0}{\frac{mM + M^2}{m}} = \frac{2Mmv_0}{mM + M^2} = \frac{2mv_0}{m + M}$$

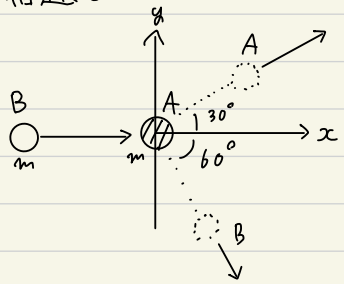
③より

$$v = \frac{mv_0 - M\left(\frac{2mv_0}{m+M}\right)}{m} = \frac{mv_0 - \frac{2Mmv_0}{m+M}}{m}$$

$$= \frac{mv_0(m+M) - 2Mmv_0}{m(m+M)} = \frac{m^2v_0 + Mmv_0 - 2Mmv_0}{m^2 + Mm}$$

$$= \frac{mv_0 - Mv_0}{m + M} = \frac{(m-M)v_0}{m+M}$$

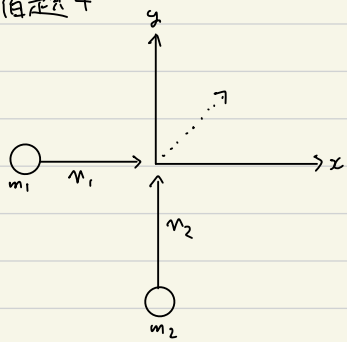
宿題3



$$v_A = v\cos 30^\circ = \frac{\sqrt{3}}{2}v$$

$$v_B = v\cos 60^\circ = \frac{1}{2}v$$

宿題4

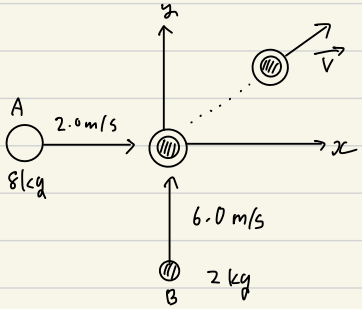


$$\begin{cases} m_1 v_1 = (m_1+m_2)v_x & \cdots ① \\ m_2 v_2 = (m_1+m_2)v_y & \cdots ② \end{cases}$$

①より $v_x = \dfrac{m_1 v_1}{m_1+m_2}$

②より $v_y = \dfrac{v_y}{m_1+m_2}$

宿題5



$$\begin{cases} 8\cdot 2 = (8+2)\vec{V}_x & \cdots ① \\ 2\cdot 6 = (8+2)\vec{V}_y & \cdots ② \end{cases}$$

①より $\vec{V}_x = \dfrac{16}{10} = \dfrac{8}{5}$

②より $\vec{V}_y = \dfrac{12}{10} = \dfrac{6}{5}$

$$\therefore \vec{V} = \sqrt{\vec{V}_x^2 + \vec{V}_y^2} = \sqrt{\frac{64}{25}+\frac{36}{25}}$$

$$= \sqrt{\frac{100}{25}} = \sqrt{4} = \underline{2\,\mathrm{m/s}}$$

$E_A = \frac{1}{2}\cdot 8\cdot 2^2 = 16$

$E_B = \frac{1}{2}\cdot 2\cdot 6^2 = 36$

$E_{AB} = 16+36 = 52$

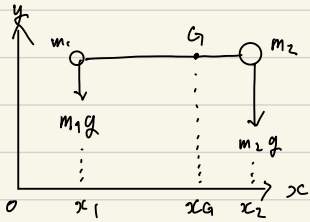
$E' = \frac{1}{2}\cdot 10\cdot 2^2 = 20$

$\therefore E = E_{AB} - E'$

$= 52 - 20$

$= \underline{36\,\mathrm{J}}$

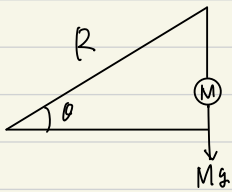
ex 2.3



$$x_G = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$\begin{aligned} N &= m_1 g (x_G - x_1) - m_2 g (x_2 - x_G) \\ &= g(m_1 x_G - m_1 x_1 - m_2 x_2 + m_2 x_G) \\ &= g(m_1 + m_2)\left(x_G - \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}\right) \\ &= g(m_1 + m_2)(x_G - x_G) \\ &= 0 \end{aligned}$$

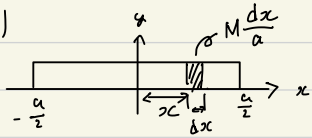
演習



$$N = -Mg R \cos\theta$$

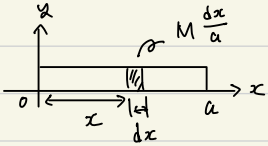
ex 3.2

(1)



$$\begin{aligned} I &= \int \sigma' r^2 \perp dl \\ &= \int_{-\frac{a}{2}}^{\frac{a}{2}} x^2 \frac{M}{a} dx = \frac{2M}{a}\int_0^{\frac{a}{2}} x^2 dx \\ &= \frac{2M}{a}\left[\frac{1}{3}x^3\right]_0^{\frac{a}{2}} = \frac{2M}{a}\cdot\frac{a^3}{24} = \underline{\frac{M}{12}a^2} \end{aligned}$$

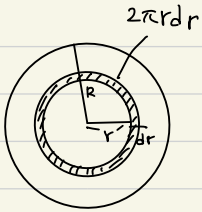
(2)



$$\begin{aligned} I &= \int \sigma' r^2 \perp dl \\ &= \int_0^a x^2 \frac{M}{a} dx = \frac{M}{a}\left[\frac{1}{3}x^3\right]_0^a = \underline{\frac{M}{3}a^2} \end{aligned}$$

ex 3.3

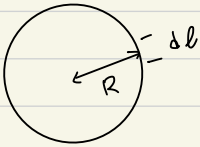
$$\sigma = \frac{M}{\pi R^2}$$



$$I = \int r^2 \perp \sigma dS = \int_0^R r^2 \frac{M}{\pi R^2} 2\pi r dr = \frac{2M}{R^2} \int_0^R r^3 dr$$

$$= \frac{2M}{R^2} \left[\frac{r^4}{4} \right]_0^R = \frac{2M}{R^2} \cdot \frac{R^4}{4} = \frac{1}{2} M R^2$$

改



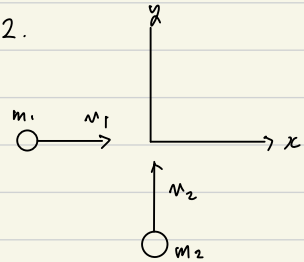
$$\sigma = \frac{M}{2\pi R}$$

$$I = \int r^2 \perp \sigma dl = \int_0^{2\pi R} R^2 \frac{M}{2\pi R} dl$$

$$= \frac{MR^2}{2\pi R} \times 2\pi R = MR^2$$

練習問題2

2.



$$\begin{cases} m_1 v_1 = (m_1 + m_2) v'_x & \cdots ① \\ m_2 v_2 = (m_1 + m_2) v'_y & \cdots ② \end{cases}$$

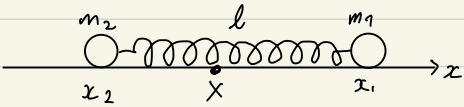
①より

$$v'_x = \frac{m_1 v_1}{m_1 + m_2}$$

②より

$$v'_y = \frac{m_2 v_2}{m_1 + m_2}$$

3.



$(x_1 > x_2)$

(1)
$$\begin{cases} m_1 \dfrac{d^2 x_1}{dt^2} = -k(x_1 - x_2 - l) & \cdots ① \\ m_2 \dfrac{d^2 x_2}{dt^2} = k(x_1 - x_2 - l) & \cdots ② \end{cases}$$

① + ②

$$(m_1 + m_2) \frac{d^2}{dt^2}\left(\frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}\right) = 0$$

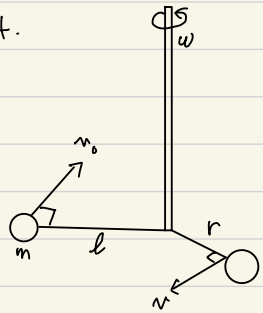
$$\therefore (m_1 + m_2) \frac{d^2 X}{dt^2} = 0$$

(2) $\frac{①}{m_1} - \frac{②}{m_2}$

$$\frac{d^2}{dt^2}(x_1 - x_2) = -\frac{k}{m_1}(x - l) - \frac{k}{m_2}(x - l)$$

$$= -k(x - l)\left(\frac{1}{m_1} + \frac{1}{m_2}\right)$$

4.

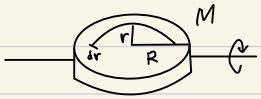


$$l m v_0 = \underbrace{r m v}_{l:\text{角運動量}}$$

$$\therefore v = \frac{l m}{r m} v_0 = \frac{l}{r} v_0$$

練習問題 3

1. (1)



$$I = \int_0^R r^2 \perp \frac{M}{\pi R^2} \pi r dr$$

$$= \frac{M}{\pi R^2} \cdot \pi \left[\frac{1}{4} r^4 \right]_0^R$$

$$= \frac{MR^4}{4}$$

$$ds = \frac{1}{2} r^2 d\theta$$